

Yue Gao

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EDUCATION

University of Wisconsin–Madison

Madison, WI

Ph.D. in Computer Sciences (advised by Prof. Kassem Fawaz)

Sep 2018 – Mar 2024

- Thesis: *Challenges and Advances in Adaptive Response Strategies for Machine Learning Security*
- Selected Courses: *Mathematical Foundations of Machine Learning, Advanced Algorithms, Non-linear Optimization.*

Shanghai University

Shanghai, China

B.S. in Computer Science and Technology (GPA 3.99/4.00, Ranked 1/292)

Sep 2014 – Jul 2018

- Thesis: *A Deep Neural Network-based Image Compression Method*
- Selected Courses: *Operating Systems, Computer Network, Assembly Language, Software Engineering.*

WORK EXPERIENCE

Research Engineer @ Google DeepMind

Mountain View, CA

Report to Chongyang Shi

May 2025 – Present

- Gemini post-training for resistance against indirect prompt injection attacks.

Product Security Engineer @ Snowflake Inc.

San Mateo, CA

Report to Sharath Sarangpur

Apr 2024 – Apr 2025

- Developed and deployed large-scale **LLM-based RAG applications** using Python, Docker, and Kubernetes.
- Evaluating the performance and adversarial robustness of LLM-based production systems for security and privacy.

Research Assistant @ University of Wisconsin–Madison

Madison, WI

Advised by Prof. Kassem Fawaz

Nov 2018 – Mar 2024

- Researched the **adversarial robustness** of multimodal ML models deployed in real-world environments.
- Systematized the security analysis of ML-based and web-based systems in **black-box** settings.

ML System Security Research Intern @ Microsoft Research

Redmond, WA

Mentored by Dr. Jay Stokes and Dr. Emre Kiciman

Jun 2021 – Sep 2021

- Proposed a research project on defenses against imperceptible textual backdoor attacks on **language models**.
- Discovered blind spots in state-of-the-art attacks and defenses, and published stronger defenses at MILCOM.
- Successfully reduced the attack success rate from 100% to 12%, even at a challenging poisoning rate of 10%.

ML Research and Development Intern @ TuCodec (Startup)

Shanghai, China

Mentored by Dr. Chunlei Cai

Jan 2018 – Jul 2018

– *ML Research*

- Secured 1st place as a **primary contributor** in the CVPR 2018 Challenge on Learned Image Compression.
- Improved the average runtime of DNN-based image compression algorithms **from 1 min to 4 secs** per 4K-res image.

– *ML Engineering & Security*

- Independently developed ML-based desktop apps on Ubuntu, MacOS, and Windows using TensorFlow and C++.
- Independently developed ML-based image compression systems using TensorFlow, Python, Docker, and Kubernetes.
- Protected ML systems from security intrusion and model stealing with security measures and anomaly analysis.

PUBLICATIONS

Conference

[1] Supply-Chain Attacks in Machine Learning Frameworks

Yue Gao, Ilia Shumailov, and Kassem Fawaz

The Eighth Annual Conference on Machine Learning and Systems (MLSys), 2025

[2] SEA: Shareable and Explainable Attribution for Query-based Black-box Attacks

Yue Gao, Ilia Shumailov, and Kassem Fawaz

Proceedings of the 3rd IEEE Conference on Secure and Trustworthy Machine Learning (SaTML), 2025

- [3] **On the Limitations of Stochastic Pre-processing Defenses**
Yue Gao, Ilia Shumailov, Kassem Fawaz, and Nicolas Papernot
Proceedings of the 36th Conference on Neural Information Processing Systems (NeurIPS), 2022
- [4] **Rethinking Image-Scaling Attacks: The Interplay Between Vulnerabilities in Machine Learning Systems**
Yue Gao, Ilia Shumailov, and Kassem Fawaz
Proceedings of the 39th International Conference on Machine Learning (ICML), 2022
- [5] **Experimental Security Analysis of the App Model in Business Collaboration Platforms**
Yunang Chen*, Yue Gao*, Nick Ceccio, Rahul Chatterjee, Kassem Fawaz, and Earlence Fernandes
31st USENIX Security Symposium (USENIX Security), 2022
- [6] **I Know Your Triggers: Defending Against Textual Backdoor Attacks With Benign Backdoor Augmentation**
Yue Gao, Jack W. Stokes, Manoj Prasad, Andrew Marshall, Kassem Fawaz, and Emre Kiciman
IEEE Military Communications Conference (MILCOM), 2022

Workshop

- [1] **Variational Autoencoder for Low Bit-rate Image Compression**
Lei Zhou*, Chunlei Cai*, Yue Gao, Sanbao Su, and Junmin Wu
Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR) Workshops, 2018

Preprints

- [1] **Human-Produced Adversarial Examples**
David Khachaturov, Yue Gao, Ilia Shumailov, Robert Mullins, Ross Anderson, and Kassem Fawaz
arXiv, 2023
- [2] **Analyzing Accuracy Loss in Randomized Smoothing Defenses**
Yue Gao*, Harrison Rosenberg*, Kassem Fawaz, Somesh Jha, and Justin Hsu
arXiv, 2020

SELECTED PROJECTS

Real-world ML Systems Security: Threat Modeling, Defending, and Characterization Sep 2020 – Jun 2023

- Investigated the security of **multi-component ML systems** exposed to diverse security threats, e.g., dependencies.
- Revealed **threats amplified by 9x** and **broke state-of-the-art defenses** by jointly exploiting multiple vulnerabilities.
- **Formally proved the non-robustness** of randomization-based defenses beyond demonstrating empirical attacks.
- Characterized the attack’s progression for **forensic purposes** and **human-explainable intelligence sharing**.

GARD: Guaranteeing AI Robustness Against Deception Jun 2019 – Feb 2024

– Leadership

- Led a 9-member cross-university team to **1st and 2nd place in grant competitions hosted by DARPA**.
- Developed initial code bases and mentored **team members with varying technical backgrounds**.
- Onboard and mentored new team members to maintain adversarial mindsets with designing defenses.

– Individual Contribution

- Performed **red teaming** and **broke over 10 internal defenses** proposed by team members prior to submission.
- Proposed **self-supervised CLIP and diffusion-based methods** to learn adversarially robust multimodal features.
- Successfully reduced the disappearance rate from 62% to 9% even **under the red-team evaluation from MITRE**.
- Contributed plug-and-play modules to the official upstream evaluation team and received acknowledgment.

Security Analysis of Online Business Collaboration Platforms Mar 2021 – Dec 2021

- Automated the analysis of security principle violations for 3K+ third-party apps in Slack and Microsoft Teams.
- Reverse engineered OAuth designs to bypass access control, and received bug bounty for medium severity.
- Demonstrated POC attacks of eavesdropping on private chats, spoofing video calls, and unauthorized code merging.

SELECTED HONORS & AWARDS

Slack Bug Bounty: Medium Severity, \$1500	2022
Top 10% Reviewers Award: NeurIPS	2022
CVPR Competition Winner: Challenge on Learned Image Compression	2018
National Scholarship: China	2017
Top 100 Elite Collegiate Award: China Computer Federation	2017
Scholarship for Exceptional Leadership: Shanghai University	2017
City Scholarship: Shanghai	2016
Outstanding Student Award: Shanghai University	2016
Outstanding Volunteer Award: ACM ICPC Asia Regional Contest	2016
Scholarship for Exceptional Innovation: Shanghai University	2016
Scholarship for Exceptional Academic Achievements: Shanghai University	2015 – 2018
Bronze Prize for Programming Contest: ACM ICPC Asia East-Continent Final Contest	2015
Bronze Prize for Programming Contest: ACM ICPC Asia Shanghai Regional Contest	2015

PROFESSIONAL ACTIVITIES

Reviewer: NeurIPS, ICML, and ICLR	2022 – 2024
External Reviewer: IEEE Symposium on Security and Privacy	2021 – 2023
External Reviewer: USENIX Security Symposium	2021 – 2022
External Reviewer: ACM Conference on Computer and Communications Security	2019
Team Leader: Collegiate ICPC Team at Shanghai University	2016 – 2017

TALKS

1. Forensics and Intelligence Sharing for ML Security <i>DARPA GARD PI Meeting, IBM Research</i>	Oct 2023
2. The Vulnerabilities of Preprocessing in Adversarial Machine Learning <i>ML Red Team, Google</i>	Oct 2023
3. The Vulnerabilities of Preprocessing in Adversarial Machine Learning <i>TrustML Young Scientist Seminar, RIKEN AIP</i>	Apr 2023
4. On the Limitations of Stochastic Pre-processing Defenses <i>DARPA GARD PI Meeting, University of Southern California (virtual)</i>	Oct 2022
5. The Interplay Between Vulnerabilities in Machine Learning Systems <i>DARPA GARD PI Meeting, University of Michigan</i>	Sep 2022
6. Experimental Security Analysis of the App Model in Business Collaboration Platforms <i>USENIX Security 2022</i>	Aug 2022
7. The Interplay Between Vulnerabilities in Machine Learning Systems <i>ICML 2022</i>	Jun 2022

TEACHING AND MENTORING

Project Mentor: DARPA GARD Project, University of Wisconsin–Madison	Fall 2023
Teaching Assistant: CS 368 (C++ for Java Programmers), University of Wisconsin–Madison	Fall 2018
Guest Lecturer: Advanced Algorithms & Data Structures, Shanghai University	2015 – 2017
Problem Designer: Undergraduate Programming Contests, Shanghai University	2015 – 2017
Student Mentor: Undergraduate Computer Science Coursework, Shanghai University	2015 – 2017

TECHNICAL SKILLS

Python	Expertise in ML and Security Research (2018 – present) and Backend Development (2016 – 2018).
PyTorch	Expertise in ML Research (2019 – present) and Distributed Training (2020 – present).
Docker	Expertise in ML Research (2018 – present) with familiarity in Kubernetes Cluster (2017 – present).
C / C++	Proficient in Security Research (2018 – 2019), Software Development (2017 - 2019), and ICPC (2014 – 2018).
Golang	Proficient in Software Engineering (2024 – present).
TensorFlow	Proficient in ML Research (2017 – 2020) and Service Deployment (2018).

ARTICLES AND MEDIA COVERAGE

CleverHans. Can stochastic pre-processing defenses protect your models?	2022
USENIX login. Experimental Security Analysis of the App Model in Business Collaboration Platforms	2022
Wired. Slack's and Teams' Lax App Security Raises Alarms	2022